Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical
Practical- 9

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Aim: User Data Management
a. Create an API to add user details using Express and MongoDB.
b. Design a React form to send user data to the backend.
c. Retrieve and display user data from MongoDB.
d. Implement an update operation for existing records.
e. Implement the delete operation for selected records.
f. Display updated records dynamically in React.

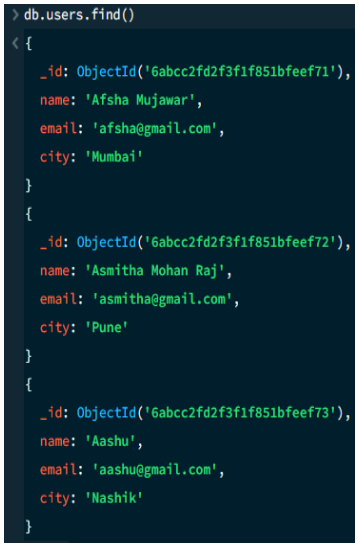
Steps:

1. Create mongo database UserDB -> collection name users
2. Find the insert all data.

Code:

```
db.users.insertMany([
{
name: "Afsha Mujawar",
email: "afsha@gmail.com",
city: "Mumbai"
},
{
name: "Asmitha Mohan Raj",
email: "asmitha@gmail.com",
city: "Pune"
},
{
name: "Aashu",
email: "aashu@gmail.com",
city: "Nashik"
},
])
```

Output:



Step 2 - Create backend

1) Create a folder practical9 & open that in vs code
2) Press ctrl+~ to open terminal & follow next step
3) Run -> npm init -y to create package.json file.
4) Add server.js file
5) Run command -> npm install mongoose express cors
6) Click + icon(new file) near practical9 & name it server.js
7) Add following code into server.js file
const express = require("express");

Code: server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// MongoDB Connection
mongoose.connect("mongodb://127.0.0.1:27017/userDB
")
 .then(() => {
 console.log("MongoDB Connected Successfully");
 })
 .catch((error) => {
 console.log("MongoDB Connection Failed");
 console.log(error);
 });
```

```
// GET - Retrieve Users
app.get("/users", async (req, res) => {
 try {
 const users = await mongoose.connection.db
 .collection("users")
 .find()
 .toArray();
 res.json(users);
 }
 catch (error) {
 res.status(500).json({message: "Error fetching users"});
 }
});
// POST - Add User
app.post("/users", async (req, res) => {
 try {
 const user = req.body;
 await mongoose.connection.db
 .collection("users")
 .insertOne(user);
 res.json({message: "User added successfully"});
 }
 catch (error) {
 res.status(500).json({message: "Error adding user"});
 }
});
// PUT - Update User
app.put("/users/:id", async (req, res) => {
 try {
 const id = new
mongoose.Types.ObjectId(req.params.id);
 const updatedUser = req.body;
 await mongoose.connection.db
 .collection("users")
 .updateOne({ _id: id },
 {
 $set: {
 name: updatedUser.name,
 email: updatedUser.email,
 city: updatedUser.city
 }
 }
 );
 res.json({message: "User updated successfully"});
 }
 catch (error) {
 res.status(500).json({message: "Error updating user"});
 }
});
// DELETE - Delete User
app.delete("/users/:id", async (req, res) => {
 try {
 const id = new
mongoose.Types.ObjectId(req.params.id);
 await mongoose.connection.db
 .collection("users")
 .deleteOne({
 _id: id
 });
 res.json({message: "User deleted successfully"});
 }
 catch (error) {
 res.status(500).json({message: "Error deleting user"});
 }
});
// Start Server
app.listen(PORT, () => {
 console.log(`Server is running on
http://localhost:${PORT}`);
});
```

Step 3 - Create frontend

1) In terminal create react project type npm create vite@latest frontend
2) Choose react, javascript, ESLint
3) In terminal move to frontend & install axios
npm install axios
4) expand src & replace App.jsx code with following

Code: App.jsx:

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
 // Store all users
 const [users, setUsers] = useState([]);
 // Form data
 const [name, setName] = useState("");
 const [email, setEmail] = useState("");
 const [city, setCity] = useState("");
 // Store ID of user being edited
 const [editId, setEditId] = useState(null);
 // Get Users from MongoDB
 const getUsers = () => {
 axios.get("http://localhost:3000/users")
 .then((response) => {
 setUsers(response.data);
 })
 .catch((error) => {
 console.log(error);
 });
 };
 // Load users when page starts
 useEffect(() => {getUsers();}, []);
 // Add or Update User
 const saveUser = () => { const user = {
 name: name,
 email: email,
 city: city
 };
 // UPDATE existing user
 if (editId)
 {axios.put(`http://localhost:3000/users/${editId}`, user)
 .then((response) => {alert(response.data.message);
 clearForm();
 getUsers();
 })
 .catch((error) => {console.log(error);});
 }
 // ADD new user
 else {axios.post("http://localhost:3000/users", user)
 .then((response) => {alert(response.data.message);
 clearForm();
 getUsers();
 })
 .catch((error) => {console.log(error);});
 }
 };
 // Edit User
 const editUser = (user) => {setEditId(user._id);
 setName(user.name);
 setEmail(user.email);
 setCity(user.city);
 };
```

```
// Delete User
const deleteUser = (id) => {
if (window.confirm("Are you sure you want to delete this user?")) {
axios.delete(`http://localhost:3000/users/${id}`)
.then((response) => {alert(response.data.message);
getUsers();
})
.catch((error) => {console.log(error);});
}
};
// Clear Form
const clearForm = () => {
setName("");
setEmail("");
setCity("");
setEditId(null);
};
return (
<div>
<h1>User Management System</h1>
{/* User Form */}
<h2>
{editId ? "Update User" : "Add User"}
</h2>
<input type="text"
placeholder="Enter Name"
value={name}
onChange={(e) => setName(e.target.value)}
/>
<br /><br />
<input type="email"
placeholder="Enter Email"
value={email}
onChange={(e) => setEmail(e.target.value)}
/>
<br /><br />
<input type="text"
placeholder="Enter City"
value={city}
onChange={(e) => setCity(e.target.value)}
/>
<br /><br />
<button onClick={saveUser}>
{editId ? "Update User" : "Add User"}
</button>
{editId && (<button onClick={clearForm}>
Cancel
</button>
)}
<hr />
{/* Display Users */}
<h2>User List</h2>
<table border="1">
<thead>
<tr>
<th>Name</th>
<th>Email</th>
<th>City</th>
<th>Actions</th>
</tr>
</thead>
<tbody>
{users.map((user) => (
<tr key={user._id}>
<td>{user.name}</td>
<td>{user.email}</td>
<td>{user.city}</td>
<td>
<button onClick={() => editUser(user)}>
Edit
</button>
<button onClick={() => deleteUser(user._id)}>
Delete
</button>
</td>
</tr>
))}
</tbody>
</table>
</div>
);
}
export default App;
```

Step 4 - Execution part

1) Split terminal
2) execute backend(server) means node server.js
3) execute frontend using npm run dev

Output:

